

Annop Sangsila

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A developer passionate about Artificial Intelligence and building software that people actually use. My journey began at Khon Kaen University and grew into teaching two core CS courses, interning on enterprise systems, and shipping software a retail store and a water factory use every day. I now focus on AI engineering RAG, agentic systems, and model fine-tuning.

EDUCATION

Khon Kaen University, Khon Kaen, Thailand

2023 - Present

B.S. (Computer Science) | GPA 3.42

SKILL & PROFICIENCIES

AI & ML: LLM APIs (tool calling, structured output), RAG & Hybrid Search, pgvector, Qdrant, Cross-Encoder Re-ranking, Hugging Face Transformers, QLoRA/PEFT, Vision LLM Extraction, Retrieval Evaluation (Recall@k, MRR)

Computer Languages: Python, TypeScript, JavaScript, Go, Java, SQL, Kotlin, PHP, C

Frontend: React, Next.js, React Native, Expo, Tailwind CSS, Flutter

Backend: FastAPI, Node.js (Express), Spring Boot, Go Fiber, Laravel

Database & Storage: PostgreSQL (pgvector), MySQL, MongoDB, Qdrant, Prisma, Firebase (Firestore), SQLite

DevOps & Tools: Git, GitHub Actions (CI/CD), Docker, Docker Compose, pytest, Postman, Figma, Vercel, Railway, Azure

WORK EXPERIENCE

Gofive

April 2026 - June 2026

Internship

Bangkok, Thailand

- Developed a multi-database architecture that isolates client data across separate databases for enterprise-scale tenants.
- Worked inside a large production codebase and its release workflow, collaborating with engineers across teams.

Todoview

June 2025 - Present

Freelance Software Developer

Remote, Thailand

- Built a custom Odoo 19 POS (Python, JavaScript/OWL, PostgreSQL 17) unifying point-of-sale, inventory, purchasing, and HR across 8 in-house modules — multi-warehouse receiving, FEFO lot and expiry handling, and PO replenishment from the stock screen with supplier cost history.
- Built daily price planning with a full change log and export pushing prices to in-store DIGI SM-100 EV label scales; added manager-only discount control, PromptPay QR payments, and Thai VAT-summary receipts.
- Covered inventory and pricing with Python unit tests and 11 Playwright E2E specs, deployed via Docker Compose across separate dev and production environments; also built the same client's water-distribution ERP — QR-scanned delivery tracking (zxing-browser) with a full audit log and role-based access across 4 roles.

Khon Kaen University

Nov 2025 - Present

Teaching Assistant

Khon Kaen, Thailand

- Selected to assist two core CS courses Object-Oriented Programming (Nov 2025 – Mar 2026) and Data Structures (Jun – Oct 2026) running lab sessions and reviewing student code with written feedback on design and correctness.
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PROJECTS

- **TaintGuard (PyPI)** Open-source Python package that blocks prompt-injected AI agent tool calls by tracking data provenance rather than scoring with a model; 215 tests, 100% coverage, mypy --strict, CI on Python 3.10–3.13.
- **Enterprise RAG & Doc Search** Document Q&A with hybrid retrieval (pgvector HNSW + Postgres full-text fused by RRF) and cross-encoder re-ranking, streaming SSE answers with line-level citations; Recall@5 0.97, MRR 0.878 on a 30-question set.
- **Market Intelligence Agent** Automated competitor analysis — a Go scraper (worker pool, rate limiting, robots.txt) feeds Qdrant, and an LLM agent writes a cited executive report with prices computed in SQL rather than generated; scheduled nightly.
- **Savy — AI Financial Agent** Expense tracker that reads Thai bank-transfer slips with a multimodal vision model into a typed Pydantic schema, plus vector RAG over spending history and an agentic budget advisor.
- **QLoRA Slip Extraction** Fine-tuning and evaluation pipeline extracting 11 fields from Thai transfer slips with a local 1.5B model, two layouts held out of training to separate reading from memorising; baseline 0.993 field accuracy on 250 slips.
- **Shabu POS — (Real-world Use)** Built and deployed a web-based POS system now in daily production use at a restaurant, handling in-store sales, table reservations, and QR-code ordering from the customer's own phone, using Next.js, TypeScript, Prisma, MySQL, and Tailwind CSS.